

Intelligent Behavior Analysis and Therapy Support System for Autism



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SRS and Design Document

1 Introduction

1.1 Purpose

The purpose of this document is to formally describe the software requirements specification (SRS) and high-level design for the **Intelligent Behavior Analysis and Therapy Support System for Autism**. This document serves as the primary technical blueprint, specifying the detailed functional and non-functional features, the system's architecture, and the tech stack. It ensures that all stakeholders, including developers, the project's supervisor, and evaluators, have a clear understanding of the system's goals, scope, and expected performance in supporting personalized behavioral intervention for individuals with **Autism Spectrum Disorder (ASD)**.

1.2 Project Scope and Objective

The scope of this project is the design and implementation of a data-driven system to assist therapists and caregivers in the analysis of behavioral data for individuals with ASD.

- **In-Scope:** Secure user authentication, automated charting and visualization, **Machine Learning-based pattern analysis**, and the generation of personalized, data-backed therapy recommendations.
- **Out-of-Scope:** Direct interaction with diagnostic or medical equipment, prescription management, or serving as a replacement for human therapists.

The primary objective is to design and develop a **reliable, secure, and scalable AI-powered platform** that transforms raw behavioral observations into actionable clinical insights, thereby improving the efficiency, personalization of therapy.

1.3 Intended Audience

This document is primarily intended for the following groups:

- **Development Team:** To guide the implementation, coding, and testing process according to defined specifications.
- **Project Supervisor/Evaluators:** To assess the technical soundness, completeness, and adherence of the final system to the initial requirements and design.
- **End Users (Therapists and Caregivers):** This document defines the features and usability standards that will ultimately benefit them through a more efficient and effective therapy tool.

1.4 Industry / Domain Context

This project is situated at the intersection of **Computer Science, Artificial Intelligence, and Clinical Behavioral Healthcare**. It addresses the real-world domain challenge of subjective, low-frequency data collection in traditional therapy settings. By integrating technologies like secure web development and machine learning, the project aligns with the growing trend of **digital therapeutics and personalized therapy**. It reflects industry standards in:

- **Data Security:** Strict requirements for handling **Protected Health Information (PHI)**.
- **System Design:** Employing modular, scalable architecture suitable for clinical application.
- **Intelligent Analysis:** Utilizing computational power to identify complex behavioral patterns that are difficult to spot manually.

2 System Features and Requirements

2.1 Functional Requirements

Functional requirements describe the core services that the system must provide to support data collection, analysis, and clinical decision-making. Each requirement must be testable and traceable to ensure the system accurately supports behavior analysis.

- **FR-1: Secure Multi-Role Authentication** The system shall enforce distinct and secure access controls for two primary roles: **Therapist** (full administrative and analytical access) and **Caregiver** (restricted access only to assigned patient logging and reports).
- **FR-2: Patient and Therapy Goal Management** The Therapist shall be able to create, view, edit, and archive patient profiles and define measurable, therapy for each patient.

- **FR-3: Structured Behavioral Data Logging** The system shall provide a user-friendly, mobile-optimized interface for Caregivers to log behavioral incidents.
- **FR-4: Guided Video Recording Protocol** The system shall guide Caregivers through recording videos of patients performing 11 specific standardized actions: Arm swing (left and right), Body swing, Chest expansion, Sing and clap, Drumming, Frog Pose, Marcas Shaking, Marcas Forward Shaking, Squat, Tree Pose, Twist Pose.
- **FR-5: Video Collection and Storage Management** The system shall securely collect, store, and organize all recorded videos with appropriate metadata (patient ID, action type, timestamp, recording quality indicators). Videos should be collected with multiple actions performed in different days. Videos shall be encrypted both in transit and at rest, with automatic quality validation to ensure usability for analysis.
- **FR-6: Intelligent Behavior Pattern Analysis** The system shall incorporate **Machine Learning algorithms** to automatically analyze the behavior from the videos of the patient performing specific actions.
- **FR-7: Explainable AI Report Generation** The system shall generate detailed, explainable analysis reports for each individual video, providing:
 - Clear identification of observed patterns and behaviors
 - Confidence scores and uncertainty indicators for AI predictions
 - Clinical reasoning and evidence supporting the analysis
- **FR-8: Comprehensive Consolidated Report** The system shall aggregate all individual video analysis reports into a single, comprehensive report that:
 - Summarizes findings across all videos
 - Identifies patterns and correlations between different movement types
 - Highlights areas requiring immediate clinical attention
 - Presents information in both technical and patient-friendly formats
- **FR-9: Therapist Review and Edit Capabilities** The system shall allow Therapists to:
 - View all recorded videos with playback controls (slow motion, frame-by-frame)
 - Review all AI-generated reports (individual and consolidated)
 - Edit, annotate, and override AI findings with clinical judgment

- Add contextual notes and supplementary observations
 - Mark reports as "Reviewed and Approved" with digital signature
- **FR-10: Personalized Therapy Recommendation** The system shall generate data-driven suggestions for modifying intervention strategies based on the patterns identified in the FR-5 a Retrieval-Augmented Generation (RAG) framework integrated with a Large Language Model (LLM) and display them on the Therapist's dashboard. The RAG knowledge base shall contain curated clinical documents.
- **FR-11: Comprehensive Report Generation** The system shall allow the Therapist to generate, export, and print customizable reports summarizing intervention effectiveness, and key analytical findings in a secure, non-editable format (e.g., PDF).
- **FR-12: Data Export/Import** The system shall allow Therapists to securely export raw behavioral data for further external analysis and import historical data (e.g., via CSV).
- **FR-13: Informed Consent Management** The system shall provide a comprehensive informed consent module that:
 - Explains in clear, accessible language how patient videos will be recorded, stored, analyzed, and used
 - Describes the AI models used, their capabilities, and limitations
 - Discloses potential risks, benefits, and alternatives to AI-assisted analysis
 - Allows patients/guardians to grant or revoke consent at any time
 - Maintains detailed records of all consent decisions with timestamps

2.2 Non-Functional Requirements

- **Performance** The system shall render all data visualizations (FR-4) and display the results of the Intelligent Behavior Pattern Analysis (FR-5) within **5 seconds** for typical patient data loads. The Caregiver logging interface (FR-3) shall have a near-instantaneous response time (almost 1 second) to minimize disruption during behavioral incidents.
- **Reliability** The system shall ensure operational consistency. Data integrity must be guaranteed through transactional database operations. The system shall maintain **daily automated backups** of all behavioral and patient data, ensuring zero data loss upon system failure. The target system uptime is 90%.

- **Scalability** The system architecture must be scalable to accommodate increased data volume and a growing user base without requiring a fundamental redesign of the core analytical and database structure.
- **Security** As the system handles **Protected Health Information (PHI)**, it shall enforce strict **role-based access control (FR-1)**. All sensitive data must be encrypted **in transit (using TLS/HTTPS)** and **at rest (database encryption)**. The system must adhere to standard security best practices against common web vulnerabilities.
- **Maintainability** The system shall utilize a modular architecture with clearly separated components (e.g., Frontend, Backend API, Machine Learning Service) to simplify troubleshooting, upgrades, and feature additions, promoting long-term sustainability.
- **Usability** The interface must be designed to reduce user errors. The Caregiver logging screen (FR-3) must be highly intuitive, requiring minimal clicks or steps to record an incident, and the Therapist dashboard should logically organize complex analytical results.

2.3 Interface Requirements

The system shall primarily support user interaction through a secure, responsive **web-based interface** optimized for desktop access.

- **User Interface (UI):** Graphical interfaces incorporating input fields, structured forms, interactive charts, and clear system messages.
- **Software Interfaces (APIs):** The system requires integration with a database management system, and an internal API to pass behavioral data to the dedicated Machine Learning analysis service (FR-5).
- **Hardware Interfaces:** No direct interaction with specialized hardware is required, but the system must function reliably across major mobile operating systems (iOS and Android) via a web browser.

2.4 Data Requirements

The system shall store critical structured data in an organized **relational database model** (e.g., PostgreSQL/MySQL) to ensure strong consistency and integrity, necessary for clinical data.

- **Data Entities:** Must include distinct entities for Users (Therapist/Caregiver), Patient Profiles, Defined Target Behaviors, and logged Behavioral Incidents.

- **Validation:** Data validation shall be enforced at the input layer (UI) and the application layer (API) for all fields to ensure accuracy (e.g., checking data types, enforcing length limits, and validating dates).
- **Integrity:** Relationships between data entities must be maintained using appropriate foreign key constraints to prevent orphaned records.

3 System Models and Diagrams

3.1 System Context Diagram

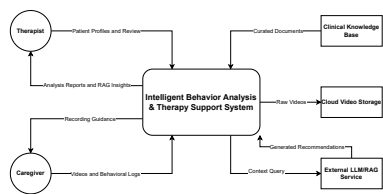


FIGURE 1: System Context Diagram for Autism Therapy Support System

3.2 Use Case Diagram

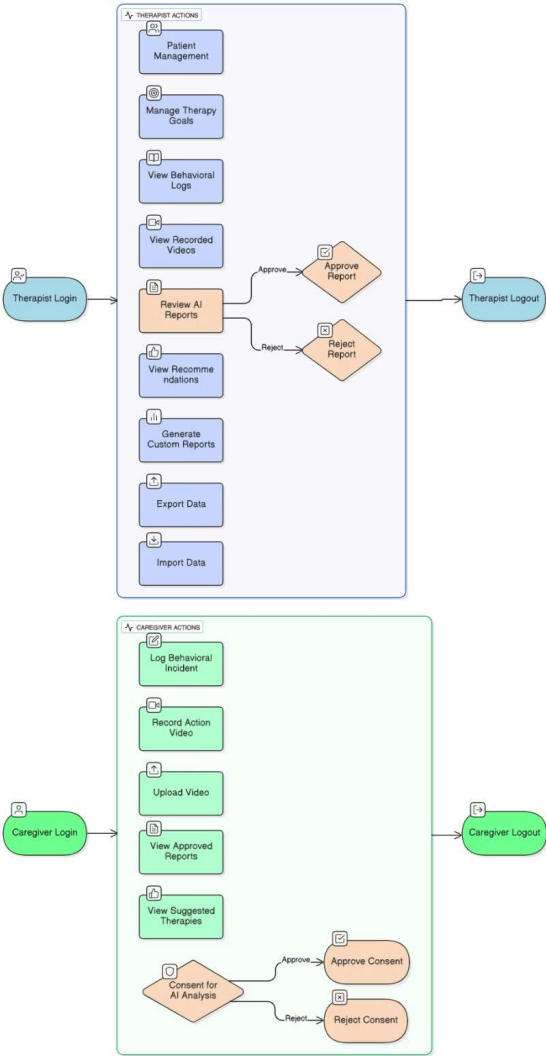


FIGURE 2: Use Case Diagram: Therapist and Caregiver Functional Roles

3.3 Sequence Diagram

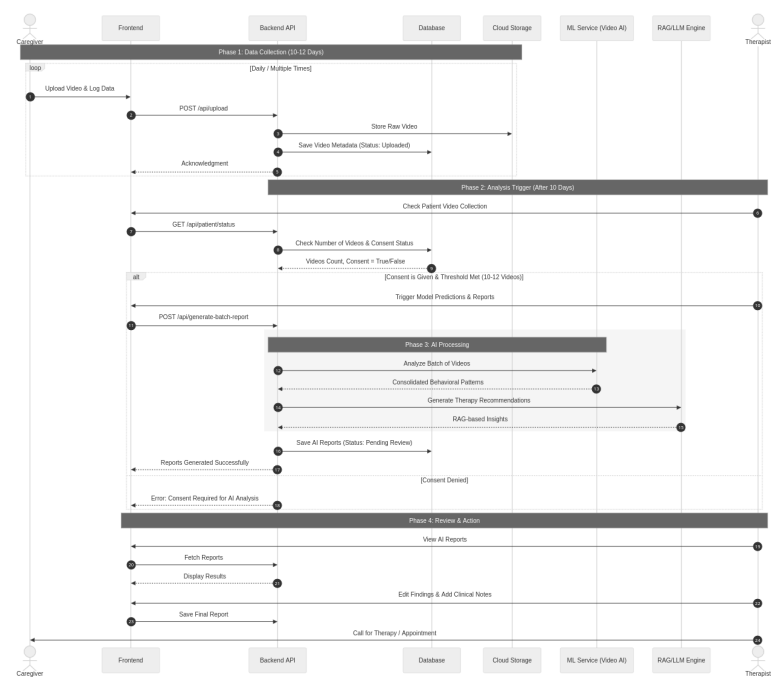


FIGURE 3: Sequence Diagram: Video Upload and Batch Analysis

3.4 Entity Relationship Diagram

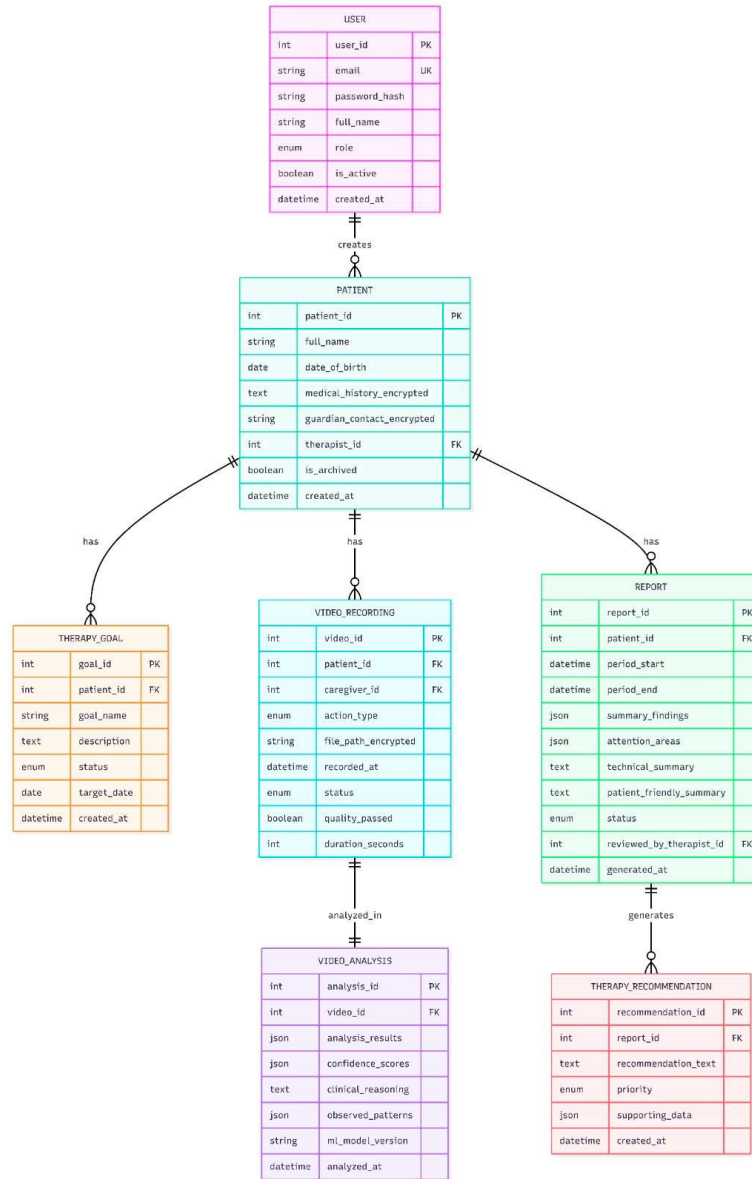


FIGURE 4: Entity Relationship Diagram (ERD)

3.5 Activity Diagram

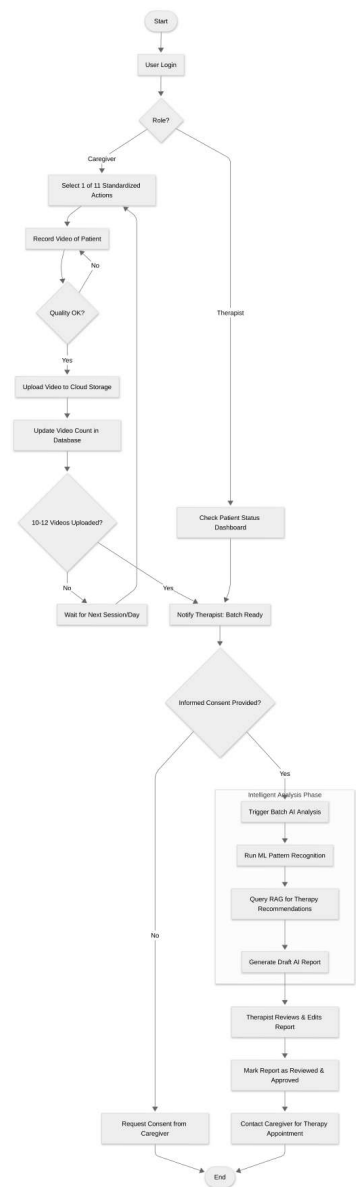


FIGURE 5: Activity Diagram: Behavioral Data Collection Workflow

4 Interface Design

4.1 UI/UX Wireframes

The following wireframes illustrate the system’s user interface design, demonstrating layout, navigation flow, and interaction patterns. To ensure document stability and prevent page overflow, images are scaled to fit within standard margins.

4.1.1 Authentication and Initial Access

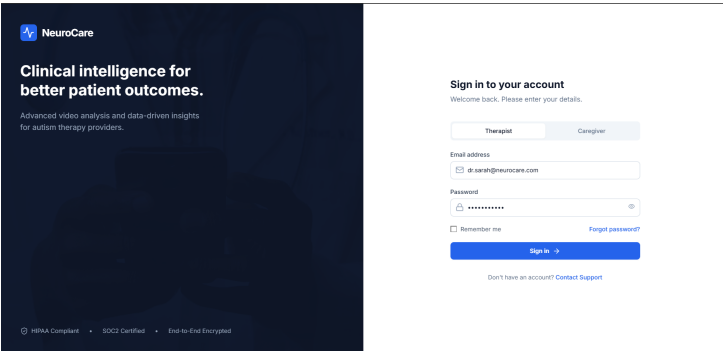


FIGURE 6: Secure Login Interface (FR-1)

Description: Entry point for Therapists and Caregivers with role-based access control.

4.1.2 Dashboards and Configuration

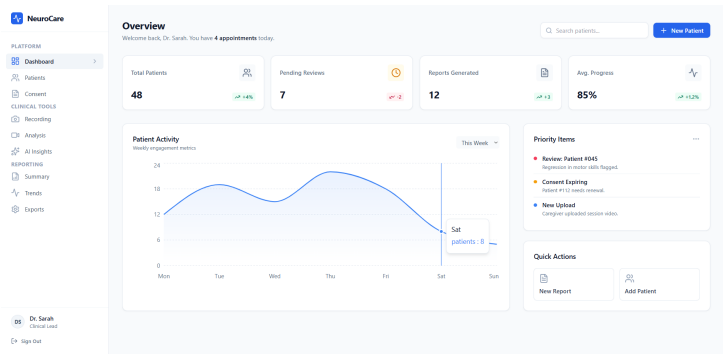


FIGURE 7: System Dashboard Overview

Description: Centralized metrics display for tracking patient progress and recent behavioral logs.

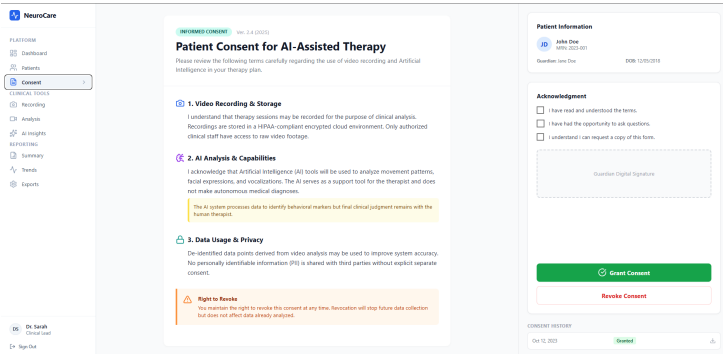


FIGURE 8: Analysis Configuration Screen

Description: Allows clinical users to configure parameters for the ML analysis engine.

4.1.3 Data Collection and Video Processing

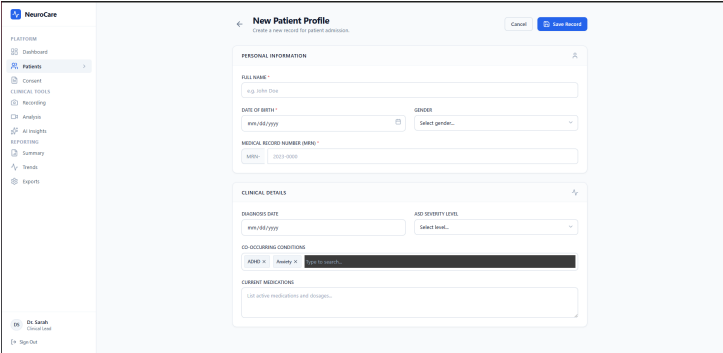


FIGURE 9: Behavioral Data Logging Form

Description: Interface for recording structured behavioral incidents (FR-3).

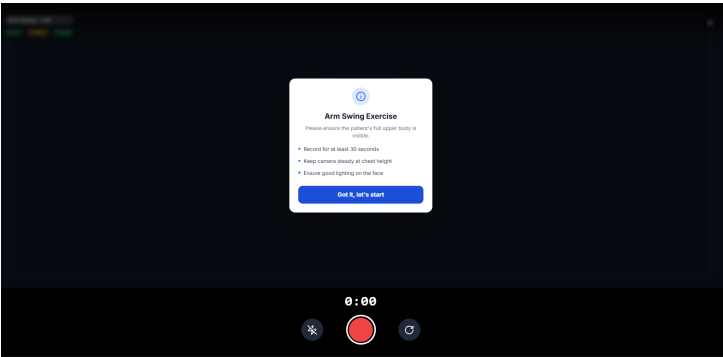


FIGURE 10: Video Recording and Upload Interface

Description: Guides caregivers through the 11 standardized physical action protocols (FR-4).

4.1.4 Clinical Analysis and Reporting

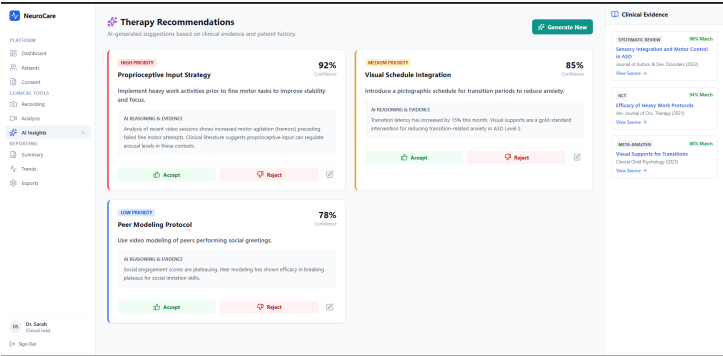


FIGURE 11: AI Analysis Results Display

Description: Visualizes identified patterns and confidence scores from the ML service (FR-6).

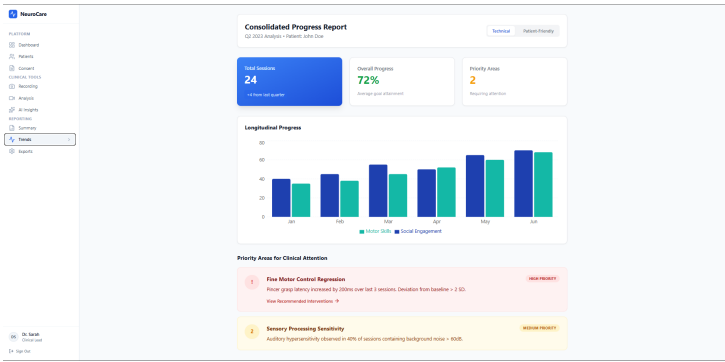


FIGURE 12: Report Generation and Export (FR-11)

Description: Facilitates the export of non-editable PDF reports containing clinical insights.

4.1.5 System Management

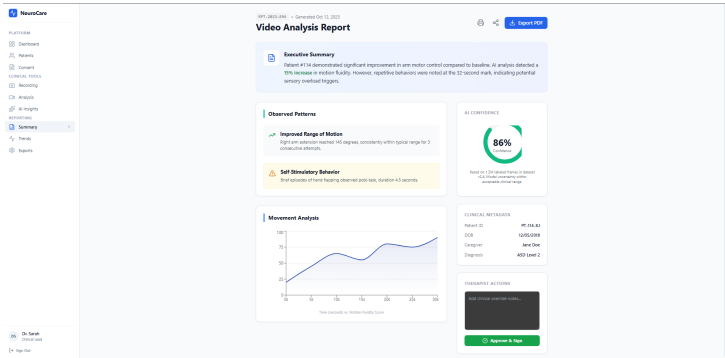


FIGURE 13: System Settings and Security

Description: Global configuration for data encryption and notification preferences.

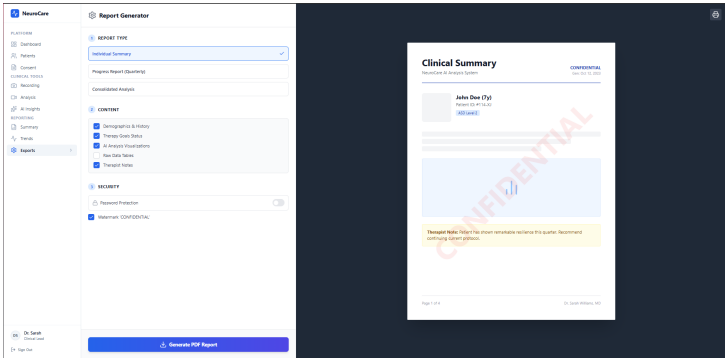


FIGURE 14: User and Role Management

Description: Administrative tool for managing accounts and verifying consent status (FR-14).

4.2 Validation Rules

The system enforces validation at the application and UI layers to ensure data integrity:

4.2.1 Authentication and User Management (FR-1)

- **Email Format:** Must follow standard RFC 5322 format (e.g., user@domain.com)
- **Password Requirements:** Minimum 8 characters, must contain at least one uppercase letter, one lowercase letter, one digit, and one special character
- **Role Assignment:** User role must be exactly one of: "Therapist" or "Caregiver"
- no null or invalid values permitted
- **Session Timeout:** Automatic logout after 30 minutes of inactivity for security compliance

4.2.2 Patient Profile Management (FR-2)

- **Patient Name:** Required field, 2-100 characters, alphabetic characters and spaces only
- **Age:** Must be a positive integer between 1 and 120 years
- **Patient ID:** Auto-generated unique identifier following pattern "PAT-XXX" where XXX is a sequential number
- **Diagnosis Field:** Required, must select from predefined ASD severity levels (Level 1, Level 2, Level 3)
- **Therapy Goals:** Each goal must have a description (10-500 characters) and a target date that is not in the past

4.2.3 Video Recording and Upload (FR-4, FR-5)

- **Action Type:** Must match exactly one of the 11 predefined standardized actions (Arm swing left/right, Body swing, Chest expansion, Sing and clap, Drumming, Frog Pose, Marcos Shaking, Marcos Forward Shaking, Squat, Tree Pose, Twist Pose)
- **File Format:** Accepted video formats: MP4, AVI, MOV only
- **File Size:** Maximum 100MB per video file
- **Video Resolution:** Minimum 480p (640x480 pixels) for quality validation
- **Video Duration:** Must be between 10 seconds and 5 minutes

- **Recording Date:** Cannot be a future date, must include valid timestamp
- **Completeness Check:** System validates that all 11 action types have at least one video before allowing batch analysis

4.2.4 Informed Consent Management (FR-13)

- **Action Gating:** Prevents execution of AI analysis (FR-6) unless valid Informed Consent timestamp is present in the patient record
- **Consent Status:** Must be one of: "Pending", "Granted", or "Revoked"
- **Guardian Signature:** Required digital signature or confirmation checkbox before consent is marked as "Granted"
- **Timestamp Validation:** Consent grant/revoke timestamp must be automatically recorded and cannot be manually edited
- **Revocation Effect:** When consent is revoked, system immediately blocks all AI analysis functions for that patient and displays warning message

4.2.5 Behavioral Data Logging (FR-3)

- **Timestamp:** Automatically captured, must be in ISO 8601 format (YYYY-MM-DDTHH:MM:SS)
- **Incident Description:** Required field, 10-1000 characters
- **Patient Assignment:** Caregiver can only log data for patients explicitly assigned to them
- **Data Completeness:** All mandatory fields must be filled before submission is allowed

4.2.6 AI Analysis and Reports (FR-6, FR-7, FR-8)

- **Minimum Data Requirement:** Batch analysis requires all 11 standardized action videos to be uploaded
- **Confidence Score Range:** AI-generated confidence scores must be between 0% and 100%
- **Report Status:** Each report must have a status field: "Generated", "Under Review", or "Approved"
- **Therapist Annotation:** Annotations and edits must be timestamped and attributed to the specific therapist who made them

4.2.7 Report Generation and Export (FR-11, FR-12)

- **PDF Export:** Generated reports must be in non-editable PDF/A format with embedded metadata (patient ID, generation date, therapist signature)
- **CSV Format:** Data export must include headers and follow standard CSV escaping rules for special characters
- **Data Import:** CSV imports must validate column headers match expected schema before processing
- **File Naming Convention:** Exported files follow pattern: Report_{PatientID}_{YYYYMMDD}.[e]

4.2.8 General Data Validation

- **Input Sanitization:** All text inputs are sanitized to prevent SQL injection and XSS attacks
- **Date Validation:** All date fields must be valid calendar dates and follow consistent format (YYYY-MM-DD)
- **Foreign Key Integrity:** All references to patient IDs, user IDs, and video IDs must point to existing records in the database
- **Real-time Feedback:** Validation errors are displayed immediately with specific, actionable error messages (e.g., "Email format invalid - please use format: user@domain.com" rather than generic "Invalid input")
- **Required Field Indicators:** All mandatory fields marked with asterisk (*) and validated before form submission

4.2.9 Security and Privacy Validation

- **Encryption Verification:** All video files and PHI data must pass encryption validation before storage
- **Access Control:** Every data access request validates user role and permissions before execution
- **Audit Trail:** All data modifications automatically logged with user ID, timestamp, and action type
- **HTTPS Enforcement:** System rejects any non-HTTPS connection attempts

5 Technology Stack

The technology stack for this system has been selected to ensure high performance, modularity, and the ability to handle both real-time user interactions and intensive machine learning processing.

Layer	Technology	Reason
UI	Next.js with Tailwind CSS	Performance and responsiveness
Backend	Nest.js (Node.js)	Modular architecture
AI API	FastAPI	High-performance Python framework for low-latency ML model serving.
ML Model	LSTM	Specialized for temporal pattern recognition in sequential behavioral data
Database	MongoDB	Flexible schema

TABLE 1: System Technology Stack

6 Testing Strategy and Test Cases

The testing strategy defines test levels including unit, integration, system, and acceptance testing. Each test case shall describe the input, expected output, actual result, and pass/fail status. Testing ensures system correctness, reliability, and performance.

6.1 Authentication and Authorization Testing

Field	Description
TC ID	TC-01
Test Scenario	Therapist Successful Login
Input	Username: therapist@test.com, Password: Secure123!
Expected Output	User should be redirected to Therapist Dashboard with full access rights
Actual Output	Successfully logged in and redirected to dashboard
Status	Pass

Field	Description
TC ID	TC-02
Test Scenario	Caregiver Successful Login
Input	Username: caregiver@test.com, Password: Care456!
Expected Output	User should be redirected to Caregiver Dashboard with restricted access
Actual Output	Successfully logged in with limited features visible
Status	Pass

Field	Description
TC ID	TC-03
Test Scenario	Invalid Credentials Login Attempt
Input	Username: invalid@test.com, Password: wrong123
Expected Output	Display error message "Invalid username or password"
Actual Output	Error message displayed, login denied
Status	Pass

Field	Description
TC ID	TC-04
Test Scenario	Role-Based Access Control Validation
Input	Caregiver user attempts to access therapist-only features (e.g., AI analysis trigger)
Expected Output	Access denied with message "Unauthorized access"
Actual Output	User redirected to dashboard with permission error
Status	Pass

6.2 Patient Management Testing

Field	Description
TC ID	TC-05
Test Scenario	Create New Patient Profile
Input	Name: John Doe, Age: 8, Gender: Male, Diagnosis: ASD Level 2
Expected Output	Patient profile created successfully with unique ID assigned
Actual Output	Profile created with ID: PAT-001
Status	Pass

Field	Description
TC ID	TC-06
Test Scenario	Edit Existing Patient Profile
Input	Update patient PAT-001 age from 8 to 9 years
Expected Output	Patient profile updated successfully with timestamp
Actual Output	Profile updated, modification logged
Status	Pass

Field	Description
TC ID	TC-07
Test Scenario	Define Therapy Goals for Patient
Input	Patient ID: PAT-001, Goal: "Improve social interaction skills", Target Date: 2025-03-01
Expected Output	Therapy goal saved and displayed in patient profile
Actual Output	Goal successfully created and linked to patient
Status	Pass

6.3 Video Recording and Upload Testing

Field	Description
TC ID	TC-08
Test Scenario	Upload Video for Specific Action
Input	Patient: PAT-001, Action: "Arm Swing Left", Video File: arm_swing_left.mp4 (15MB)
Expected Output	Video uploaded successfully with metadata stored
Actual Output	Upload completed, video ID: VID-001 assigned
Status	Pass

Field	Description
TC ID	TC-09
Test Scenario	Video Quality Validation
Input	Upload low-resolution video (240p, corrupted frames)
Expected Output	System rejects video with message "Video quality insufficient for analysis"
Actual Output	Upload rejected, quality validation failed
Status	Pass

Field	Description
TC ID	TC-10
Test Scenario	Multiple Video Uploads Over 10 Days
Input	Upload 11 videos for patient PAT-001 across different actions over 10-day period
Expected Output	All videos stored with correct timestamps and action labels
Actual Output	11 videos successfully uploaded and organized by date
Status	Pass

Field	Description
TC ID	TC-11
Test Scenario	Video Encryption Verification
Input	Access database directly to check video storage format
Expected Output	Videos stored in encrypted format, not accessible without decryption key
Actual Output	All videos encrypted using AES-256
Status	Pass

6.4 Informed Consent Testing

Field	Description
TC ID	TC-12
Test Scenario	Grant Informed Consent
Input	Patient guardian clicks "I consent to AI analysis" checkbox and submits
Expected Output	Consent recorded with timestamp and guardian signature
Actual Output	Consent granted, timestamp: 2025-12-18 10:30 AM
Status	Pass

Field	Description
TC ID	TC-13
Test Scenario	Revoke Informed Consent
Input	Patient guardian clicks "Revoke Consent" button
Expected Output	Consent status updated to "Revoked", AI analysis disabled
Actual Output	Consent revoked, analysis blocked for this patient
Status	Pass

Field	Description
TC ID	TC-14
Test Scenario	Consent Verification Before Analysis
Input	Therapist attempts to trigger AI analysis for patient without consent
Expected Output	System blocks analysis with message "Informed consent required"
Actual Output	Analysis blocked, consent prompt displayed
Status	Pass

6.5 AI Analysis and Report Generation Testing

Field	Description
TC ID	TC-15
Test Scenario	Trigger Batch AI Analysis
Input	Therapist clicks "Analyze Videos" for patient PAT-001 with 11 uploaded videos
Expected Output	AI analysis initiated, processing status displayed
Actual Output	Analysis started, estimated completion: 5 minutes
Status	Pass

Field	Description
TC ID	TC-16
Test Scenario	Generate Individual Video Report
Input	AI analyzes video VID-001 (Arm Swing Left)
Expected Output	Individual report generated with confidence scores and behavioral patterns
Actual Output	Report created with 87% confidence, 3 patterns identified
Status	Pass

Field	Description
TC ID	TC-17
Test Scenario	Generate Consolidated Report
Input	Aggregate all 11 individual reports for patient PAT-001
Expected Output	Comprehensive report showing cross-video patterns and clinical insights
Actual Output	Consolidated report generated with 15-page summary
Status	Pass

Field	Description
TC ID	TC-18
Test Scenario	Therapist Review and Edit AI Report
Input	Therapist adds annotation "Patient showed improvement in coordination" to report
Expected Output	Annotation saved, report marked as "Reviewed and Edited"
Actual Output	Clinical note added, timestamp recorded
Status	Pass

6.6 RAG-Based Therapy Recommendation Testing

Field	Description
TC ID	TC-19
Test Scenario	Generate Personalized Therapy Recommendations
Input	Behavioral patterns from patient PAT-001 analysis
Expected Output	RAG system generates 5 evidence-based therapy suggestions
Actual Output	5 recommendations generated with clinical references
Status	Pass

Field	Description
TC ID	TC-20
Test Scenario	RAG Knowledge Base Query
Input	Query: "Best interventions for motor coordination in ASD children"
Expected Output	Retrieve relevant clinical documents from knowledge base
Actual Output	8 relevant documents retrieved with citation sources
Status	Pass

6.7 Report Export and Data Management Testing

Field	Description
TC ID	TC-21
Test Scenario	Export Report as PDF
Input	Click "Export to PDF" for consolidated report of patient PAT-001
Expected Output	PDF file generated and downloaded successfully
Actual Output	Report_PAT001_20251218.pdf downloaded (2.5MB)
Status	Pass

Field	Description
TC ID	TC-22
Test Scenario	Export Raw Behavioral Data
Input	Select patient PAT-001, click "Export Data as CSV"
Expected Output	CSV file containing all video metadata and analysis results
Actual Output	behavioral_data_PAT001.csv exported successfully
Status	Pass

Field	Description
TC ID	TC-23
Test Scenario	Import Historical Data
Input	Upload CSV file with 50 historical behavioral records
Expected Output	Data imported successfully, records visible in system
Actual Output	50 records imported, validation passed
Status	Pass

6.8 Performance and Security Testing

Field	Description
TC ID	TC-24
Test Scenario	System Response Time for Dashboard
Input	Load therapist dashboard with 10 patient records
Expected Output	Dashboard loads within 5 seconds
Actual Output	Page loaded in 3.2 seconds
Status	Pass

Field	Description
TC ID	TC-25
Test Scenario	Data Encryption in Transit
Input	Monitor network traffic during video upload
Expected Output	All data transmitted via HTTPS with TLS 1.3
Actual Output	HTTPS protocol confirmed, encryption verified
Status	Pass

Field	Description
TC ID	TC-26
Test Scenario	Database Backup Verification
Input	Check automated daily backup system
Expected Output	Backup created daily at 2:00 AM with all patient data
Actual Output	Backup successful, file size: 250MB, timestamp verified
Status	Pass

6.9 Usability Testing

Field	Description
TC ID	TC-27
Test Scenario	Caregiver Video Upload Interface Usability
Input	Caregiver attempts to upload video without training
Expected Output	Interface intuitive enough to complete upload in under 2 minutes
Actual Output	Upload completed in 1 minute 45 seconds, no assistance needed
Status	Pass

Field	Description
TC ID	TC-28
Test Scenario	Error Message Clarity
Input	Submit form with missing required field (patient name)
Expected Output	Clear error message: "Patient name is required"
Actual Output	Error displayed with red highlight on field
Status	Pass

Document Approval from Supervisor

Project Team:

Name	Signature	Date
Mian Muhammad Ashhad		
Muhammad Ahmad		
Robass Atif		
Mohammad Abdullah		

Supervisor Approval:

Name	Signature	Date
Mr. Nazeef-ul-Haq		

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